

Three Lenses in Cybersecurity: Networking • Security • Information Assurance

Networking

- Networking is how computers talk to each other.
- Devices connect using cables (Ethernet) or wireless signals (Wi-Fi).
- Networks allow sharing files, sending messages, accessing websites, and connecting to servers.
- Routers direct traffic so data reaches the right device.
- IP addresses act like home addresses for computers.
- Without networking, the internet wouldn't exist.

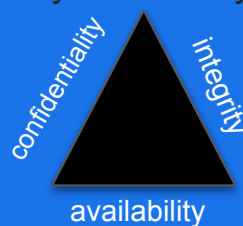
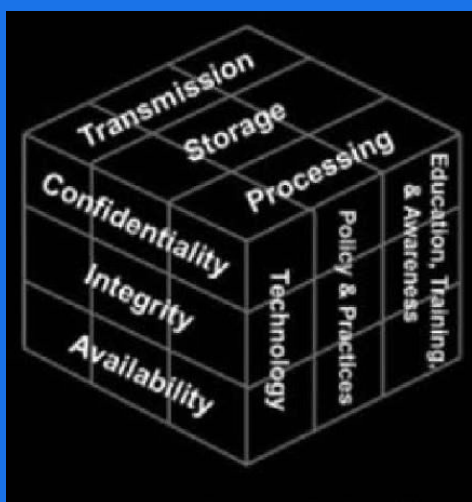
Security

- Focuses on keeping systems safe from anything that can break in, break out, or break things.
- Uses tools like firewalls, encryption, monitoring, and access controls.
- Thinks in real time: What threats exist right now, and how do we stop them?
- Makes sure attackers can't steal, damage, or mess with your systems.
- Very technical, very active the lock the doors and watch the cameras part of cybersecurity.

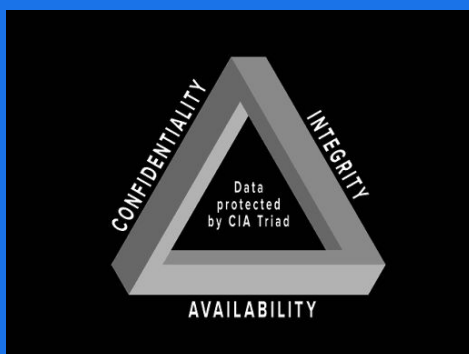
Information Assurance

- Uses policies, audits, compliance checks, and risk management.
- Thinks long term, How do we manage and protect data over its entire life?
- Ensures data isn't corrupted, lost, or mismanaged.
- More governance focused the "set the rules and make sure everyone follows them" part.

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cube



What is the CIA triad?



Threat → Vulnerability →
Risk → Control Flow
Arrow

